

TRINITY-1 — STATUS

Remnant Fieldworks Inc. · Coherent Inheritance Framework (CIF) · ExecutionProof-governed

As of: hardware run **COMPLETE**, independently verified, aggregate verdict **ALLOW**.

Result (live IBM Heron r2 hardware)

One preregistered Bell-CHSH nonclassicality witness, evaluated independently on three distinct quantum processors and fused into a single chain-linked ExecutionProof record.

Slot	Backend	S	$n\sigma$	Verdict	Job
D1	ibm_kingston	2.7360	23.27	VALID_ABOVE	d9e46l-cjeosc73fiaj00
D2	ibm_fez	2.5785	18.29	VALID_ABOVE	d9e46p-cjeosc73fiaj50
D3	ibm_marrakesh	2.7410	23.43	VALID_ABOVE	d9e4a5sin-v1c73apq1pg

- Classical bound $|S| \leq 2.0 \cdot \text{Tsierson } 2\sqrt{2} \approx 2.828 \cdot \text{certification threshold } S \geq 2.2$.
- Aggregate ExecutionProof verdict: **ALLOW** · trinity_certified = True
- TRINITY nonce: 508f0ad692a7bd3a1d1a069aa02f4dc2519560153f646220f2150bbe47270e2f
- Record hash: 8d23accac81791fa10f1dba1be79a132168966c4620fc42d16de656bcf9d688b
- Chain — previous record (OMNI-1): 22ffb2ba137c89c2846225c6d5a11fb2a4dc6aaa6a0ba1dfe6864b-c50393bb0c
- NIST beacon pulse: 1865761 @ 2026-07-19T03:21:00.000Z
- Shots: 4,000 per circuit × 4 circuits × 3 devices = **48,000** total (matches prereg + locked harness --shots default; OMNI shot-count-consistency lesson honored).
- **Independent verifier: PASS** (recomputes every S, verdict, witness hash, nonce, record hash, and the aggregate verdict from raw counts — no qiskit).

Completed

- [x] **Preregistration** (TRINITY-1-preregistration.md) — three-processor design, identical CHSH witness, per-device + aggregate verdict logic (ALLOW/HOLD/DENY/GATE-STOP), fused trinity_nonce , kill condition, hardware gate, scope/honesty caveats, purpose gate.
- [x] **Harness** (trinity_harness.py) — builds the CHSH witness, runs three independent devices (simulator or hardware), assembles the trinity-proofrecord-1.0 record, computes the aggregate ExecutionProof verdict, writes report + JSON. Records the **actual physical qubits** used after transpilation per circuit (post-layout).

- [x] **Independent verifier** (`trinity_verify.py`) — reconstructs every device `S`, verdict, `device_witness_hash`, `trinity_nonce`, and `record_hash` from raw counts, no qiskit.
- [x] **Preregistration + harness lock** (`MANIFEST.sha256`) — SHA-256, re-verified `OK`.
- [x] **Simulator validation** — three independent noisy aer backends:
- **NOMINAL** → all three `VALID_ABOVE` → **ALLOW**; verifier **PASS**.
- **HOLD** scenario → device 3 driven to `INCONCLUSIVE` → aggregate **HOLD**.
- **DENY** scenario → device 3 driven below classical (`INVALID`) → aggregate **DENY**.
- **Tamper test** → altering one raw count without recomputing the hash → verifier recomputes **DENY** and reports **FAIL** (`record_hash` mismatch caught).
- **Manifest-lock test** → editing the harness after locking broke the declared hash → provenance invalid → **DENY** (the lock works).
- [x] **Hardware run** on `ibm_kingston` + `ibm_fez` + `ibm_marrakesh` — genuine CHSH violation on all three, aggregate **ALLOW**, verifier **PASS**.

Scope & honesty (unchanged from preregistration)

- This is a **device-dependent** demonstration of nonclassicality reproduced across three processors and cryptographically fused into one governance record. It is **not** device-independent, does **not** close the detection or locality loopholes, and does **not** entangle the three processors with one another. Each device is a separate, self-contained CHSH experiment; TRINITY is the reproduction + fusion claim, nothing more.

Provenance / reproducibility

```
cd /home/ubuntu/trinity-testbeds
sha256sum -c MANIFEST.sha256      # confirm prereg + harness lock intact
python3 trinity_verify.py         # independent reconstruction from raw counts →
PASS
```

Security note

The IBM Quantum token used for this run was pasted in chat and is therefore considered **compromised** — rotate it at <https://quantum.cloud.ibm.com>. The harness only reads the token from the `IBM_QUANTUM_TOKEN` environment variable — never hard-coded, never committed (`.gitignore` excludes `.env`).